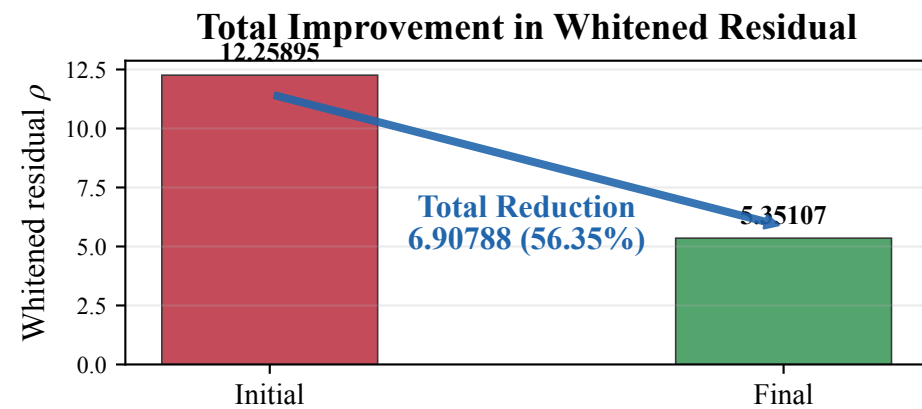
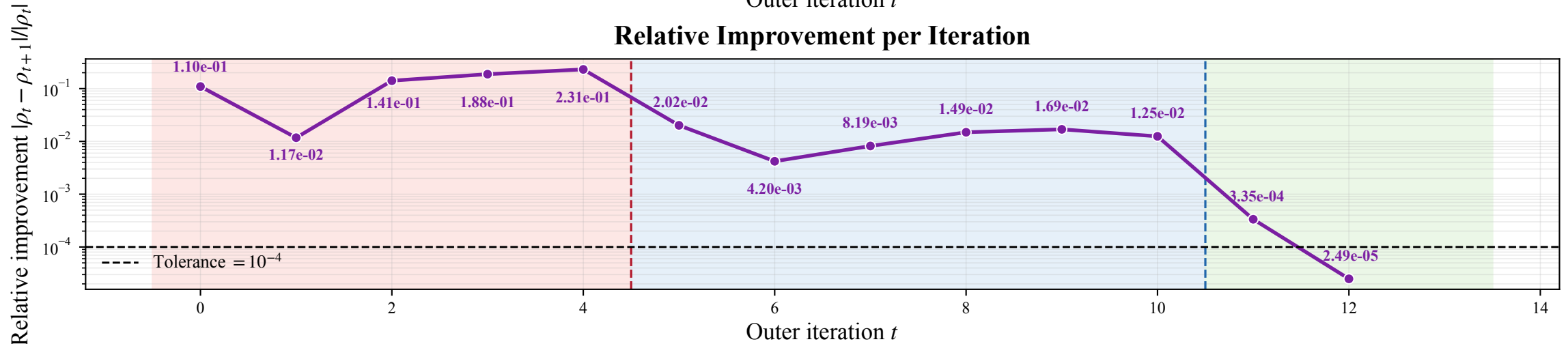
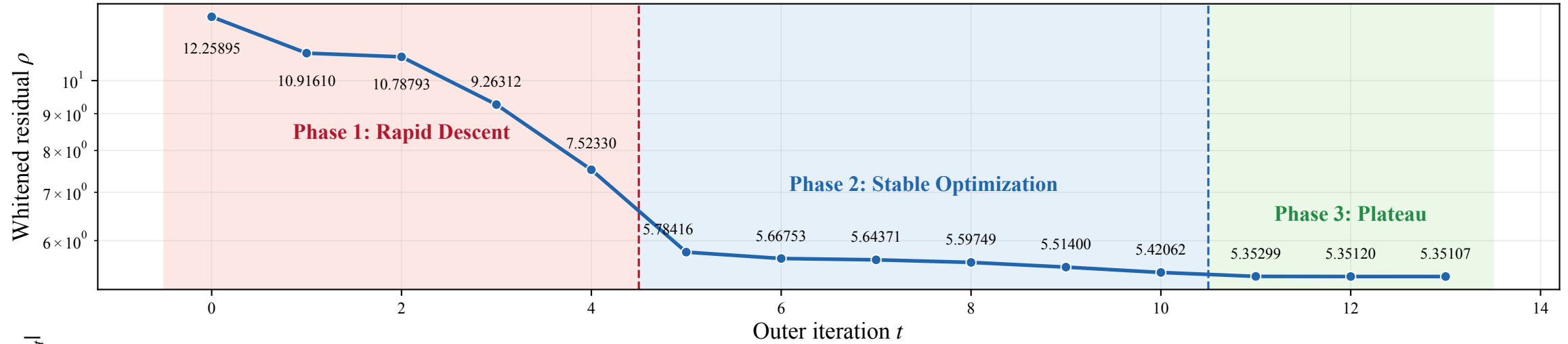


Algorithm 2 Convergence Analysis

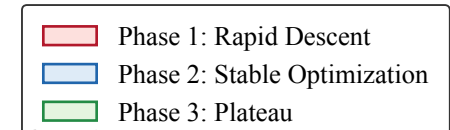
Trust-Region-Style Safeguarded Multi-Pass MFG Batch Refinement

Convergence of Whitened Residual ρ



Algorithm 2 Performance Summary

Metric	Value
Number of outer iterations	13
Initial residual ρ_{init}	12.258951
Final residual ρ_{final}	5.351066
Total reduction	6.907885 (56.35%)
Stop reason	small relative improvement
Final rotation error (rad)	1.896824e-01
Final translation error	9.494125e-04



Phase 1
 $t=0$ to 4
Large residual reduction

Phase 2
 $t=5$ to 10
Stable trust-region refinement

Phase 3
 $t=11$ to final
Marginal improvement / plateau

Note: Relative improvement is defined as $|\rho_t - \rho_{t+1}|/|\rho_t|$. The convergence threshold is $1e-04$.